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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/696,402	03/28/2024	Koji TAKEUCHI	Q296691	2585

23373 7590 09/17/2025

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EXAMINER	
HUYNH, LUAT T	

ART UNIT	PAPER NUMBER
3667	

NOTIFICATION DATE	DELIVERY MODE
09/17/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/696,402	Applicant(s) TAKEUCHI et al.	
	Examiner Luke Huynh	Art Unit 3667	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 03/28/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-4 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1 and 3 is/are rejected.

8) ☑ Claim(s) 2 and 4 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 03/28/2024, 12/18/2024.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA. Claims 1-4 have been examined

Priority

2. Acknowledgment is made of applicant's claim for foreign priority under 35 U.S.C. 119 (a)-(d).

Information Disclosure Statement

3. The information disclosure statement (IDS) submitted on 03/28/2024 and 12/18/2024 is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Claim Rejections - 35 USC § 102

4. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

5. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

6. Claims 1 and 3 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Buckley et al. (US 2008/0218381 A1) (Buckley hereinafter).

7. Regarding claim 1, Buckley discloses a vehicle-exit assist apparatus, comprising:

a target information acquisition device (12) configured to detect a target existing on a rear side of an own vehicle, and to acquire, as target information, information on the detected target (Fig. 2, sensors 14 and 16), and

a control unit (10) configured to execute vehicle-exit assist control of assisting an occupant of the own vehicle in safely exiting the own vehicle (Fig. 1, control module 12),

wherein the control unit is configured to:

determine, based on the target information, whether an interference target which is liable to interfere with the safe exit of the occupant from the own vehicle during a vehicle stop is detected; determine whether a vehicle-exit assist condition is satisfied, the vehicle-exit assist condition being satisfied at least when the interference target is detected; determine, when the vehicle-exit assist condition is satisfied, whether a specific condition is satisfied, the specific condition being a condition to be satisfied when at least a part of the interference target exists in a rear-side vehicle width region being a region between a virtual line extending from a left rear corner portion of the own vehicle toward a rear direction in a front-and-rear direction of the own vehicle and a virtual line extending from a right rear corner portion of the own vehicle toward the rear direction; execute the vehicle-exit assist control when the specific condition is not satisfied; and avoid executing the vehicle-exit assist control when the specific condition is satisfied ([0022] – [0028]).

8. Regarding claim 3, Buckley discloses the vehicle-exit assist apparatus according to claim 1, as stated above, wherein the control unit (10) is configured to determine that the specific condition is satisfied further when a component of a speed of the interference target in the front-and-rear direction of the own vehicle is higher than 0 and is lower than a

predetermined speed threshold value and/or a component of a deceleration of the interference target in the front-and-rear direction is equal to or higher than a predetermined deceleration threshold value ([0022]).

Allowable Subject Matter

9. Claims 2 and 4 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

10. The following is a statement of reasons for the indication of allowable subject matter: claim 2 recites, inter alia, "of a front end portion of the interference target, a portion closest to the own vehicle in a vehicle width direction of the own vehicle is defined as a proximate portion, virtual lines that each have a predetermined length, are positioned on a reference axis passing through the left rear corner portion and the right rear corner portion of the own vehicle, and extend toward vehicle width outward directions from the left rear corner portion and the right rear corner portions, respectively, are defined as intersecting determination lines; and the control unit is configured to: determine that the detected target is an interference target when an intersection between an extension line extending from the proximate portion along a moving direction of the detected target and the reference axis is positioned on one of the intersecting determination lines, and a predicted period predicted to be required for the detected target to reach the intersection is equal to or shorter than a predetermined period threshold value; and determine that the specific condition is satisfied further when a magnitude of a change amount per unit period of a position of the intersection is equal to or larger than a predetermined change amount threshold value". The art of record does not disclose the above limitations, nor would it be obvious to modify the art of record so as to include the above limitations.

Prior Art

11. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. See attached form PTO-892.

Conclusion

12. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Luke Huynh whose telephone number is 571-270-5746. The examiner can normally be reached Mon 8-5, Tues 8-12, Thurs & Fri 8-2.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Hitesh Patel can be reached at 571-270-5442. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/LUKE HUYNH/
Examiner, Art Unit 3667

09/15/2025